

The entry lemma of the pivot band

Flank interlacing, a source-defined $n_0 = \lfloor 2N/5 \rfloor$, and the L^2 remainder that is half-filling

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Abstract

Round 380 proved that a rank-one Loewner chain cannot raise negative inertia: an entry $\text{ind}_-(A_{n_0}) \leq 1$ at a source-defined index n_0 is transported through the whole band (`adaptive_band_from_entry`). Equivalently, by the Jacobi count, the monic pivots $h_0, \dots, h_{n_0}$ of the signed source Hankel have at most one sign change. The strongest form is $n_0 = N - 1$ with zero changes — the free-window floor plate, which is lemma L^* itself. The three-term recurrence alone does not forbid a second prefix return (a matched scramble dies at degree 21, with 37 negative pivots in the prefix). This note isolates the source ingredient that does: every negative run on the ordered cosine support has length at most 2, and the run's ν -mass is at most two-thirds of the immediately flanking μ -mass. Under that condition the two-period adversary of amplitude $2/3$ first goes negative at $\sim N/2$, and the reduced entry $n_0 = \lfloor 2N/5 \rfloor$ sits strictly below that kill and strictly above the scramble death. What remains for $n_0 = N - 1$ is an L^2 restriction estimate for the Christoffel–Darboux kernel on the flank blocks — the same coherence remainder that L^* is. Nothing in this document is evidence for or against the Riemann hypothesis.

Status. Lemmas 1–3 are proved (finite RKHS and Hankel algebra). Lemmas 4–7 are proved (exact five-, three-, six- and seven-atom identities). Lemma 6 is proved (two-period $c \geq 1$ kills h_0 or H_1). Proposition 2 is a finite machine census of 85 MAIN windows, six terminal-dead χ -sprouts, and a matched scramble, not a cofinal law. Proposition 3 is a finite table of two-period first negatives, not a theorem of every amplitude. The original statement — a source-defined $n_0 \leq N - 1$ with $\text{ind}_- H_{n_0+1} \leq 1$ on every canonical window — is *reduced* to the flank condition of §7 plus the named L^2 remainder of §8. No RH claim. No L^* claim.

1 The object

Let $X = \{x_j = \cos(\pi j/S) : j = 1, \dots, S\}$ with $S = 2N - 1$ odd, and let w be a real atomic weight on X (the folded signed von-Mangoldt tent density times the mesh factor $1 - x$). Write μ for the positive part and ν for the negative part, so $w = \mu - \nu$ as signed measures on the same grid, with disjoint supports. Write $H_r(\rho)$ for the $r \times r$ Hankel determinant of a weight ρ , $H_0 := 1$, and

$$h_j(\rho) = \frac{H_{j+1}(\rho)}{H_j(\rho)}$$

for the monic Jacobi norms whenever the minors are nonzero. The μ -orthonormal Christoffel–Darboux kernel of depth k is K_k^μ , and E_k is that kernel dressed by ν (the Gram BB^T of [1, 2]). The identity

$$\lambda_{\max}(E_k) < 1 \quad \Longleftrightarrow \quad H_k(w) \succ 0 \quad \Longleftrightarrow \quad h_0(w), \dots, h_{k-1}(w) > 0$$

is the finite dictionary of lemma L^* at depth k .

Round 380 records two kernel theorems, used here as black boxes. **rankOne_inertia_antitone**: $\text{ind}_-(A+vv^T) \leq \text{ind}_-(A)$. **adaptive_band_from_entry**: if $A_{n+1} = A_n + v_n v_n^T$ and $\text{ind}_-(A_{n_0}) \leq 1$, then $\text{ind}_-(A_n) \leq 1$ for every $n \geq n_0$. The Jacobi count **indNeg_hankel_eq_neg_pivot_count** identifies $\text{ind}_- H_{n_0+1}$ with the number of negative prefixes among $h_0, \dots, h_{n_0}$. The entry lemma is therefore a statement about those prefixes, at a source-defined n_0 .

2 What the recurrence sees

The scaled signed Stieltjes chain on (X, w) produces the Jacobi coefficients α_k, β_k^2 with $\beta_k^2 = h_k/h_{k-1}$. A sign change of h is exactly a negative β^2 .

Proposition 1 (Recursion, census). *On the flagship window $z = 16$ ($N = 184$) the source chain has $|\alpha_k| \leq 0.57$ through degree $N/2$ and the first $|\alpha| > 2$ occurs at the first negative pivot $k = N$. The matched scramble (seed 1) stays mild through degree 20 and explodes at the first negative pivot ($\alpha_{21} = +3.23$), with $\beta_{20}^2 < 0$ and $\lambda_{\max}(E_{22}) > 1$. The first dangerous approach $h_k \rightarrow 0$ is not a source/scramble separator: both worlds drop eight log-units of $|h|$ by degree 6.*

The α -explosion is a *symptom* of the first sign change, not a cause that could be bounded a priori from the three-term coefficients alone. Leg A therefore kills the hope that a sign/size profile of (α, β^2) without the atomic support would distinguish the source: any bound that uses only the recurrence until it breaks is a restatement of the first negative pivot.

3 Finite identities

Lemma 1 (RKHS sum). *For every real polynomial p of degree $< k$,*

$$\int p^2 d\nu \leq \left(\sum_j \nu_j K_k^\mu(y_j, y_j) \right) \int p^2 d\mu.$$

Consequently $\sum_j \nu_j K_k^\mu(y_j, y_j) < 1$ implies $H_k(w) \succ 0$.

Proof. $p(y) = \langle p, K_k^\mu(\cdot, y) \rangle_\mu$ on the span of degree $< k$, so $p(y)^2 \leq K_k^\mu(y, y) \int p^2 d\mu$ by Cauchy-Schwarz in $L^2(\mu)$. Sum against ν . $\square$

The sum bound is SATZ and source-defined. It is also coarse: on the flagship window the largest k with the sum strictly less than one is $K_{\text{sum}} = 9 = 0.049 N$, and the scramble already has sum 1.78 at $k = 9$. It does not yield $K/N \rightarrow \kappa > 0$.

Lemma 2 (Pair energy). *If $m_r = \int x^r dw$, then*

$$m_0 m_2 - m_1^2 = \frac{1}{2} \sum_{i,j} w_i w_j (x_i - x_j)^2.$$

In particular $h_1 > 0$ whenever $m_0 > 0$ and the right-hand side is positive. On the five-atom interlacing toy of Lemma 4 both sides equal 56.

Proof. Expand the right-hand side: $\sum_{i,j} w_i w_j (x_i^2 - 2x_i x_j + x_j^2) = 2m_0 m_2 - 2m_1^2$. $\square$

Lemma 3 (Christoffel comparison). *Let p run over monic polynomials of degree k . Then*

$$h_k(w) = \min_p \int p^2 d(\mu - \nu) \geq (1 - \lambda_{\max}(E_{k+1})) h_k(\mu),$$

whenever $\lambda_{\max}(E_{k+1})$ is defined. Since $h_k(\mu) > 0$, the sign of $h_k(w)$ is decided by whether E_{k+1} is a strict contraction — the L^ dictionary at depth $k+1$. On the three-atom flank toy of Lemma 5 one has $h_1(w) = 6 \geq (1 - \frac{1}{3}) \cdot 6 = 4$.*

Proof. Degree k is degree $< k+1$, so $\int p^2 d\nu \leq \lambda_{\max}(E_{k+1}) \int p^2 d\mu$. Take the μ -minimizer of the monic class. $\square$

4 Interlacing toys

Lemma 4 (Five-atom 2-versus-1). *On $\{-2, -1, 0, 1, 2\}$ with weights $\{2, -1, 2, -1, 2\}$ one has $h_0 = 4$, $h_1 = 14$, $h_2 = 13$, $h_3 = -144/7$. Here $S = 5$, $N = 3$, $S_+ = 3$, $S_- = 2$, and $h_0, h_1, h_2 > 0$ is the full free window of the toy. The first negative pivot is post-cap.*

Lemma 5 (Three-atom flank). *On $\{0, 1, 2\}$ with weights $\{3, -2, 3\}$ the unique negative run has flanking ratio $2/(3+3) = 1/3$, and $h_0 = 4$, $h_1 = 6$, $h_2 = -3$. Here $N = 2$, and $h_0, h_1 > 0$ is again the free window. The one-atom kernel identity $\nu(\{1\})K_2^\mu(1, 1) = 1/3 = \lambda_{\max}(E_2)$ is exact.*

Lemma 6 (Equal two-period kills H_1). *On $\{0, 1, 2, 3, 4, 5\}$ with weights $\{+1, -1\}^3$ one has $H_1 = 0$. Total mass vanishes, so $h_0 = 0$.*

Lemma 7 (Clustered run of three). *On $\{0, \dots, 6\}$ with weights $\{2, 2, -1, -1, -1, 2, 2\}$ one has $H_1 > 0$, $H_2 > 0$, $H_3 < 0$. Here $N = 4$, and the first negative pivot sits at degree $2 < N - 1$. A negative run of length 3 is therefore not forbidden from killing the free window by the moment count alone.*

5 The flank condition

Order the cosine nodes as a path. A *negative run* is a maximal consecutive block of ν -atoms.

Definition 1 (Flank). *A signed atomic measure on a linearly ordered support is c -flanked if every negative run has length at most 2 and the ν -mass of the run is at most c times the sum of the at most two immediately adjacent μ -atoms. Write (F1) for the run-length bound, (F2) for c -flank with $c < 1$, and $(F2_{2/3})$ for $c = 2/3$.*

The constant $2/3$ is the first rational strictly above every core-42 flanking ratio (all ≤ 0.619). It is not fitted to the extension anchors; seven deep EXT windows have $2/3 < r_{\max} < 1$ and are excluded from $(F2_{2/3})$ while still satisfying (F1) and (F2). The 3/8-floor and MED-CAP of the gap process (r361) enter only as τ -separation of the nodes: consecutive cosine gaps are comparable, so a run of length ≤ 2 is a $O(1/S)$ block in θ . They are not themselves a pivot theorem.

Proposition 2 (Flank census). *On the sealed 85-window MAIN ladder: (F1) holds 85/85, (F2) holds 85/85 (every $r_{\max} < 1$, no unflanked run), and $(F2_{2/3})$ holds 78/85. The seven EXT-heavy windows are $kz \in \{69, 96, 97, 99, 107, 117, 129\}$, with worst $r_{\max} = 0.879$ at $kz = 129$. On the core 42, no run has ratio $\geq 2/3$, and at most 4.02% of runs have ratio $\geq 1/2$. The six terminal-dead χ -sprouts ($chi3$ 15, 19, 23, 33, 39 and $chi4$ 20) all satisfy (F1) and $(F2_{2/3})$; terminal death is not a prefix-pivot failure. The matched scramble fails (F1) (max-run 5, nine runs of length ≥ 3) and (F2) ($r_{\max} = 2.71$, twelve runs with ratio ≥ 1), and its first negative pivot is $21 < \lfloor 2 \cdot 184/5 \rfloor = 73$.*

Uniform four-node blocks with $\nu/\mu < 1$ do *not* hold on MAIN (flagship overlapping $r_{\max} = 2.08$). The source structure is the *run* against its flanks, not a dyadic partition of the grid. Digamma/tent regularity of the weights is used only through this discrete comparison: the same cosine mesh with scrambled signs produces long runs and flank ratios above one.

6 Two-period amplitudes

Proposition 3 (Two-period table). *On the cosine grid of $S = 81$ nodes ($N = 41$), the signed two-period $w_j = (1 - x_j)(\pi/S) \cdot (+1 \text{ even } j, -c \text{ odd } j)$ has first negative pivot*

c	1/2	2/3	4/5	1
first neg.	40	20	10	0

so $\text{first}/N \approx 0.98, 0.49, 0.24, 0$. The same ratios, to two digits, reproduce at $S = 21, 41, 161$. Amplitude $c = 2/3$ is therefore a half-filling adversary for a theorem that uses only (F1) and (F $_{2/3}$): it forbids claiming $n_0 = N - 1$, and it permits $n_0 = \lfloor 2N/5 \rfloor = 16$ on this row ($20 \geq 16$). Amplitude $c \geq 1$ kills at h_0 , matching Lemma 6.

A two-period of amplitude 0.88 (the EXT-heavy $r_{\max}$) dies at $\sim 0.12N$, which is the scramble wall. That is why a uniform- c theorem covering the seven EXT-heavy windows cannot beat the generic bound, and why (F $_{2/3}$) is the honest sufficient condition: it excludes those seven from the $\kappa = 2/5$ claim while still covering the core 42 and the dead χ . The seven remain under (F1)+(F2) with $c < 1$, which is enough for h_0, h_1 and for the RKHS-sum entry, not for a positive fraction of N .

7 The reduction

Definition 2 (Source-defined entry). *On a window of half-filling N set*

$$n_0(w) = \lfloor \frac{2}{5}N \rfloor$$

whenever w is (2/3)-flanked in the sense of Definition 1, and declare the entry failed otherwise. This n_0 is a function of the ordered signed weight alone — no spectral input, no first-negative readout.

Proposition 4 (Entry on the census). *On every MAIN window, $h_k > 0$ for all $k \leq n_0(w)$ (in fact through $k = N - 1$, the r377 free-window census). On the scramble, $n_0 = 73$ and the first negative pivot is $21 < 73$, so the scramble is excluded by the premise of Definition 2 and by the conclusion. On every two-period row of Proposition 3 at $c = 2/3$, the first negative pivot is at least n_0 .*

The reduced lemma is therefore:

*If w is (2/3)-flanked, then $h_0, \dots, h_{n_0} > 0$ with $n_0 = \lfloor 2N/5 \rfloor$. Equivalently $\text{ind}_- H_{n_0+1}(w) = 0 \leq 1$. By **adaptive_band_from_entry** this entry activates the rank-one band from n_0 .*

The implication from (F1)+(F $_{2/3}$) to the prefix bound is SATZ on the toys of §4, on the two-period family at $c = 2/3$ (Proposition 3), and as a census on the construction. A cofinal L^2 proof of the same implication is the named remainder of the next section.

8 What half-filling still needs

The RKHS sum (Lemma 1) gives $K/N \rightarrow 0$. An $\|p\|_\infty$ plus Bernstein-in- θ estimate grows like $\sqrt{S}$ and likewise cannot yield a uniform positive fraction of N . The bound that *would* give $K/N \rightarrow \kappa > 0$ is an operator-norm estimate on flank blocks: if E_k were localized at scale S/k , then $\lambda_{\max}(E_k) \leq c + \text{tail}_k$ with $c \leq 2/3$, and tail_k small for $k \leq (2/5)N$. At $k = N$ the same estimate becomes $\lambda_{\max}(E_N) < 1$, which is L^* .

Thus:

- $n_0 = \lfloor 2N/5 \rfloor$ under (F $_{2/3}$) is the reduced entry (strictly smaller than L^* , strictly larger than the scramble death $0.114N$, strictly below the two-period kill $\sim N/2$).
- $n_0 = N - 1$ with zero sign changes is L^* and is *not* a consequence of (F1)+(F $_{2/3}$) alone: the two-period of amplitude $2/3$ is a named adversary against that claim.

- The missing input for half-filling is the L^2 Christoffel tail on the flank blocks (r285 coherence, the assist of $\lambda_{\max} = \text{maxdiag} \cdot (1 + \text{assist})$), together with whatever stronger source regularity (the $\delta_{1/2} \leq 0.04$ quantile of the core, the tent profile) upgrades $c = 2/3$ toward $c = 1/2$, where the two-period table reaches $n_0 = N - 2$.

9 Kill battery

world	(F1)	(F2 _{2/3})	first neg.
MAIN 85	85/85	78/85	$\geq N$ (r377)
core 42	42/42	42/42	$\geq N$
dead χ (6)	6/6	6/6	$\geq n_0$
scramble w9	fail (run 5)	fail ($r_{\max} = 2.71$)	$21 < 73$
two-period $c = 2/3$	pass	pass	$\sim N/2 \geq n_0$
two-period $c = 1$	pass	fail	0
clustered run 6	fail	—	$< n_0$

Scramble drops at both premises of the sufficient condition (named: run length and flank ratio). Dead χ satisfy both; (F2_{2/3}) is not the terminal-death mechanism. The two-period of amplitude 2/3 is the binding adversary against raising κ from 2/5 to 1, and of amplitude 1 against dropping the $c < 1$ clause.

10 Lean docking (named Prop, not a sorry)

No Lean file is edited in this round. The docking statement for R384, against the r380 kernel in `RH/PivotCoordinate.lean`, is the following named Prop. It is not a `sorry`: the same convention as `ComplementaryDualHankelInertia / CauchyInterlace`.

```
def FlankEntryPrefix : Prop :=
  ∀ {S : Nat} (m : SignedAtoms S),
  FlankTwoThird m →
  let N := (S+1)/2
  let n0 := (2*N)/5
  (∀ i ≤ n0, 0 < m.h i)
```

Here $(2 \cdot N)/5$ is Lean natural division, hence $\lfloor 2N/5 \rfloor$. The previous real-number form $2N/5 \leq n_0$ would have been false at non-multiples of 5.

`FlankTwoThird` is Definition 1 with $c = 2/3$ on the ordered support of $\mathbf{m}$. Given that `Prop`, `indNeg_hankel_eq_neg_pivot_count` turns the prefix into `ind_` $H_{n_0+1} = 0$, and `adaptive.band.from.entry` transports the bound along any Loewner chain issuing from that entry. The Prop does not assert (P1) or (P2) on a window, does not assert L^* , and does not mention the Riemann hypothesis.

Claim boundary

Finite identities, a named sufficient condition, a construction census, and a reduced entry $n_0 = \lfloor 2N/5 \rfloor$. Not a theorem of L^* , not a theorem of the wall, and not evidence for or against the Riemann hypothesis in either direction.

References

- [1] TFPT RH program, round 283: full-source quasi-definiteness and the L^* reduction.
- [2] TFPT RH program, round 285: Christoffel decomposition $\lambda_{\max} = \text{maxdiag} \cdot (1 + \text{assist})$.
- [3] S. Hamann, *Post-cap pivots of the signed source*, `rh/problem/postcap_pivots.pdf`, 2026.
- [4] `RH/PivotCoordinate.lean`, round 380: rank-one inertia and adaptive band, sorry-free.